Project Gameplan

Use it for every project

1. **Plan on paper (5 min)**
   Write in plain English what the program does, step by step, before writing any code. For a bigger project, also fill in the Roadmap Template on the next page.

2. **Build it blind**
   Code it without looking at the lesson or old files. Stay stuck at least 5 minutes before you peek.

3. **Small steps**
   Write a few lines, run, check. Never write the whole thing before running it.

4. **Break it on purpose**
   Try bad inputs and typos, read the errors, and fix them.

5. **Explain it out loud**
   Go line by line like you're teaching an apprentice. Any line you can't explain is what to review.

6. **Send it to Practice Coach and Socratic Tutor**
   Get hints on fixes and dig into the why behind each piece, not full answers.

7. **Rebuild tomorrow**
   Rewrite it from a blank file, then compare.

Ready to move on?

I can build the day's project from scratch without peeking.
I can explain every line.

If not, stay on that day. Going slower is fine.

Roadmap Template

Fill this in on paper before a bigger project, then follow it so you never build at random

Part A. Draw the flowchart

- Start with a rounded box that says Start and end with one that says Stop.
- Every action is a box (deal, add, print).
- Every yes/no question is a diamond with a labeled arrow for each answer.
- Anything that repeats gets an arrow that loops back up to the question.
- Check: can you trace every path from Start to Stop with your finger?

Part B. Split it into stages

Each stage is one small piece that runs and can be tested by itself. A good order:

1. Setup: the data and starting values.
2. One pass with no choices: the simplest version that runs start to finish.
3. Add one decision at a time.
4. Add the repeating part (loops).
5. Special cases and edge rules.
6. Finish and restart.
7. Tidy up: remove repeated code and break it on purpose.

Part C. Fill in one card per stage

Stage name	
Goal (one sentence)	
Hints as questions (no code) **What tool does this? Where does it live?**	
Test it **How will I know it works?**	
Done?	[ ] Runs [ ] Tested [ ] Can explain it

Roadmap rules

- One stage at a time. Run and test it before you start the next.
- If something breaks, look in the stage you just wrote first.
- Stuck for 5 minutes? Write down the exact question and ask Practice Coach or Socratic Tutor for a hint.
- Never skip ahead to polish before the simple version works.